

WORKING PROTOTYPE · OFFLINE-FIRST · WINDOWS ON SNAPDRAGON

FieldLens Zero

A no-cloud crisis intelligence system for Snapdragon-powered HP PCs.

FieldLens Zero turns chaotic photos, voice notes, and text from a disaster zone into a verified, privacy-protected action packet — entirely on a Snapdragon-powered HP PC, even when the network is down.

— THE GAP

Emergency AI assumes the one thing disasters remove.

Cloud-first tools

Need connectivity, expose sensitive evidence, and return free-form answers that are difficult to transfer or audit.

FieldLens Zero

Runs locally, separates evidence from inference, locks high-risk actions for review, and exports small signed packets.

The overlooked problem

Emergency-response information rarely arrives as clean database rows. It arrives as a shaky photo, a voice note in a local language, a partial location, or a message relayed through several people. During floods, fires, landslides, and network outages, cloud-dependent AI can fail precisely when it is needed most. Responders then spend critical time translating, retyping, checking privacy, and deciding which report is most urgent.

FieldLens Zero treats connectivity failure as the starting condition, not an edge case.

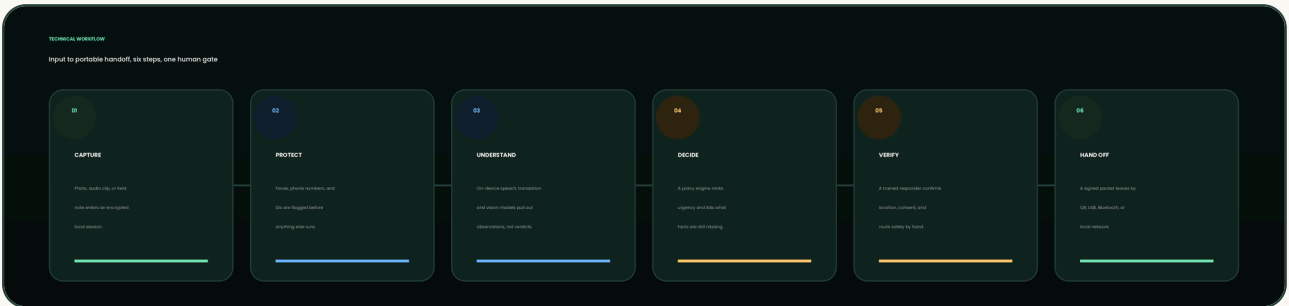
— THE PROPOSED SOLUTION

A portable, offline incident-intelligence station.

A Snapdragon-powered HP PC becomes the station. A responder adds a photo, an audio clip, and a field note — nothing more specialised than what they already carry. From there, the local AI pipeline runs the full chain on-device:

- 1 Transcribes and translates multilingual audio.
- 2 Extracts observable facts from images and text.
- 3 Separates evidence from inference.
- 4 Detects urgent medical, shelter, route, and safety needs.
- 5 Redacts personally identifiable information.
- 6 Highlights uncertainty and missing facts.
- 7 Proposes an action sequence without automatically dispatching it.
- 8 Exports a compact, tamper-evident handoff packet for transfer by QR, USB, Bluetooth, or local network.

Every recommendation is linked to supporting evidence. High-risk actions remain locked until a human confirms location, consent, and route safety.



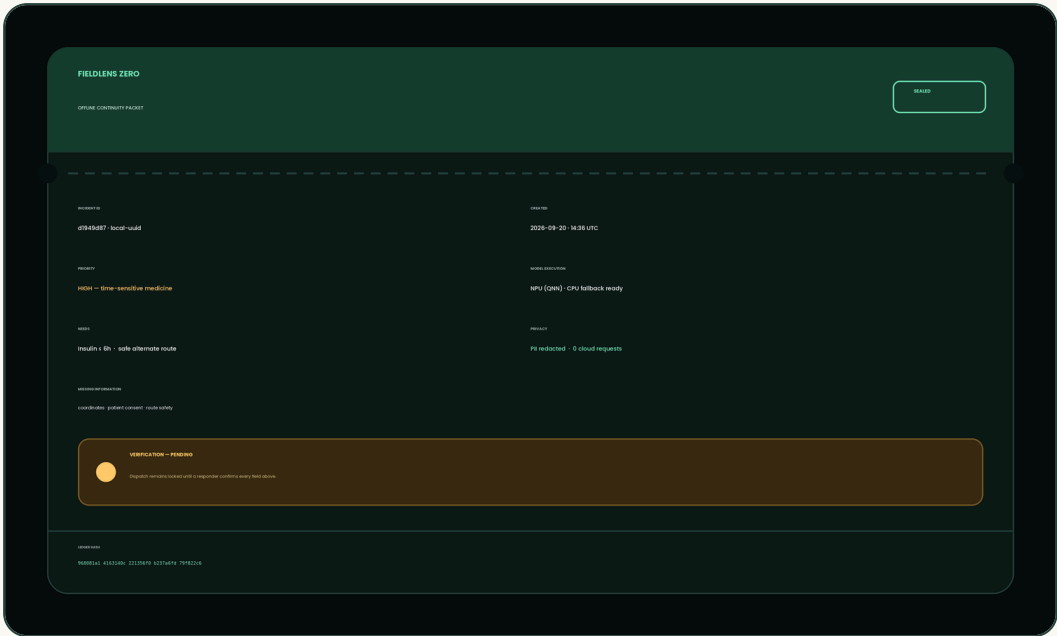
Six macro-stages of the on-device pipeline, from capture to a verified, portable handoff.

CORE INNOVATION

Continuity packets, not another chatbot.

The key output is an **offline continuity packet**. It contains a structured incident summary, evidence hashes, urgency, requested resources, translation, privacy state, uncertainty, and human-verification status. The packet is small enough to move through low-bandwidth or physically carried channels. When connectivity returns, verified packets can be synchronised without uploading raw sensitive evidence by default.

This design makes local AI operationally useful rather than merely conversational.



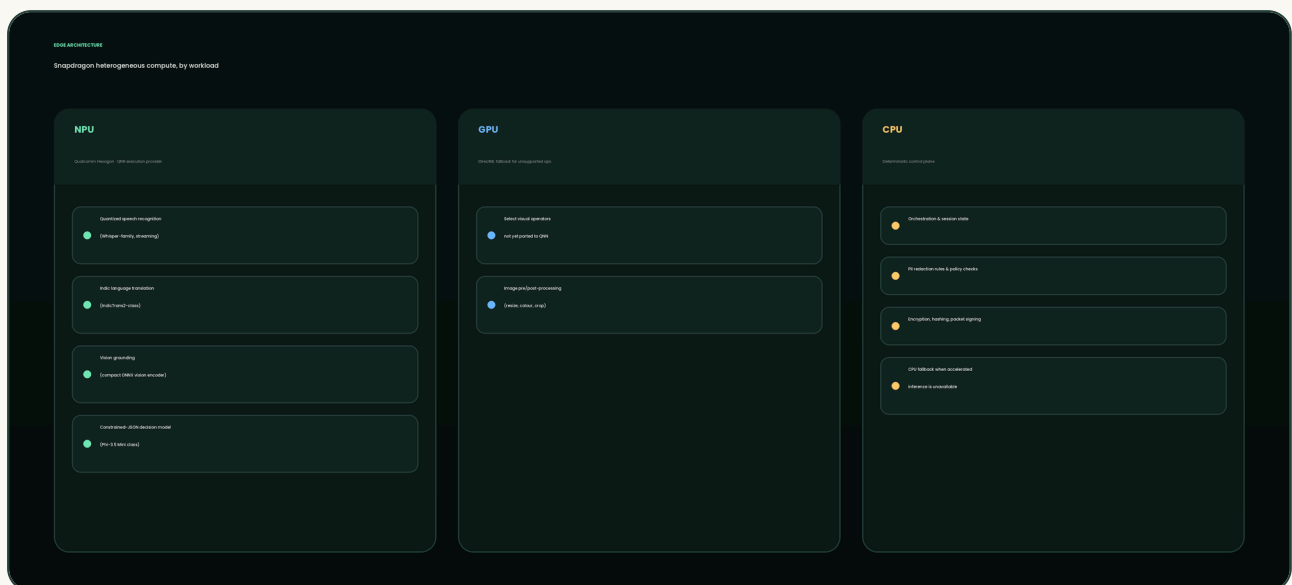
A sample continuity packet, generated from the bundled demo scenario — sealed, redacted, and awaiting human sign-off.

WHY SNAPDRAGON-POWERED HP PCS

The workload is sustained, multimodal, and privacy-sensitive.

That profile is well matched to heterogeneous on-device compute — one reason the design leans on the NPU first rather than treating it as an afterthought.

- **NPU** — quantized speech, language, and vision inference.
- **GPU** — selected visual operators.
- **CPU** — orchestration, policy checks, packet signing, and deterministic redaction.
- Local execution reduces cloud dependency, network latency, and sensitive-data exposure.
- A laptop adds battery operation, a large review surface, local storage, keyboard input, and familiar field deployment.



Workload placement across Snapdragon's heterogeneous compute.

The intended Windows-on-Snapdragon build uses ONNX Runtime with the Qualcomm QNN execution provider, quantized models, static shapes where practical, cached sessions, and adaptive inference profiles based on battery, thermals, and urgency.

— AI MODEL PLAN

Modular by design, so no single model blocks the release.

Built to use models available through **Qualcomm AI Hub** where suitable, or compatible open-source alternatives:

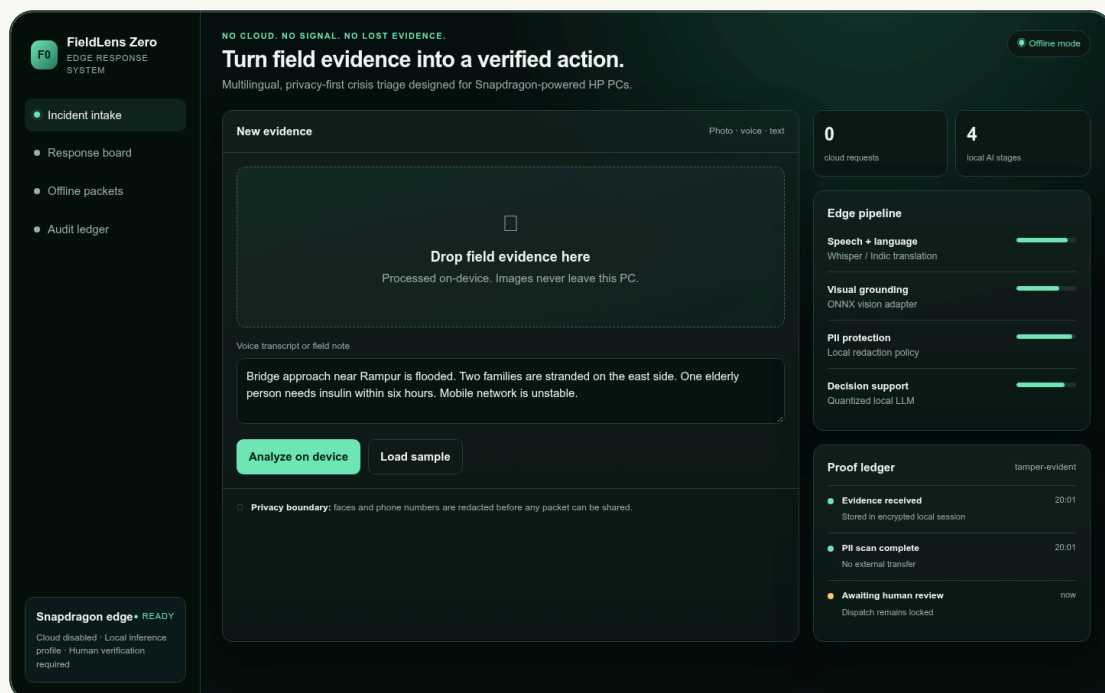
- A quantized Whisper-family model for speech recognition.
- An Indic language translation model such as IndicTrans2.
- An ONNX-compatible compact vision model such as Florence-2 or MobileCLIP class.
- A compact instruction model such as Phi-3.5 Mini class for constrained JSON summarisation.
- OpenCV and local NER / pattern rules for face and PII redaction.

The model list is modular, so an unsupported operator can be swapped out without changing the incident-packet format. Final selection rests on licence review, QNN compatibility, and accuracy, latency, memory, and energy tests on the target HP PC.

— WORKING PROTOTYPE

Designed for decisions, not conversation.

The interface exposes urgency, evidence, missing facts, privacy state, and model activity. High-risk output stays behind a human-verification gate.



The browser-based prototype: incident intake, local edge pipeline status, and the tamper-evident proof ledger.

— TRUST & SAFETY

Deliberately not autonomous.

- Never dispatches medical or rescue resources by itself.
- Labels AI-generated content and exposes confidence.
- Distinguishes observed facts from inferred claims.
- Blocks export while critical high-risk fields are unverified.
- Keeps an append-only proof ledger of model version, edits, approvals, and packet hashes.
- Supports deletion and retention policies for sensitive local evidence.
- Falls back to CPU and a manual workflow when a model is unavailable.

— EVALUATION PLAN

A multilingual, consented scenario set — floods, landslides, medical access, shelter, damaged infrastructure.

Accuracy & safety

Fact-extraction precision/recall · urgent-need detection recall · translation adequacy reviewed by speakers · PII redaction recall · unsupported-claim and human-correction rates · % of high-risk exports correctly blocked.

Snapdragon performance

Cold- and warm-start latency · end-to-end time per incident · peak memory and energy per incident · sustained performance and thermals · NPU/GPU/CPU utilisation · offline reliability over a multi-hour run.

Human factors

Time from evidence receipt to verified packet · task completion and error recovery · responder trust calibration · keyboard-only usability and low-bandwidth handoff.

— ACCESSIBILITY & DEPLOYMENT

Built for a trained volunteer, not a power user.

Large controls, keyboard navigation, high contrast, and clear risk labels. Language support is modular. The app is planned as an ARM64 Windows package with an offline installer, a local model bundle, and no mandatory sign-in. A browser prototype demonstrates the interaction without network access.

TARGET OUTCOMES — TO VALIDATE ON THE TARGET DEVICE, NOT CURRENT CLAIMS

0 CLOUD REQUESTS	<20s BALANCED-MODE RESULT	100% HIGH-RISK PACKETS GATED
≥95% PII RECALL TARGET	4h+ MIXED-USE BATTERY TARGET	p95 LATENCY REPORTED

ROADMAP

- 1 Working prototype** — interface, schema, packet export, verification gate, and proof ledger.
- 2 Snapdragon integration** — export and quantize models, test QNN coverage, implement execution routing, and benchmark on a Snapdragon-powered HP PC.
- 3 Field validation** — consented multilingual scenarios, red-team tests, policy refinement, and human-correction measurement.
- 4 Pilot** — offline ARM64 installer and a supervised pilot with a disaster-management, health, campus-safety, or humanitarian partner.

EXPECTED IMPACT

FieldLens Zero can shorten the path from messy evidence to a verified response while protecting the people represented in that evidence. The continuity-packet design can later support rural health camps, campus safety, humanitarian logistics, and industrial emergency teams.

CLOSING

The project does not ask, “How can a laptop run a chatbot?” It asks, “What capability becomes possible when a private, efficient AI operations room can travel anywhere?” Snapdragon-powered HP PCs make that answer practical: critical information remains useful even when the cloud disappears.

01 A continuity packet — not another chatbot.	02 Privacy and human review are built in.	03 Snapdragon makes offline operations practical.
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